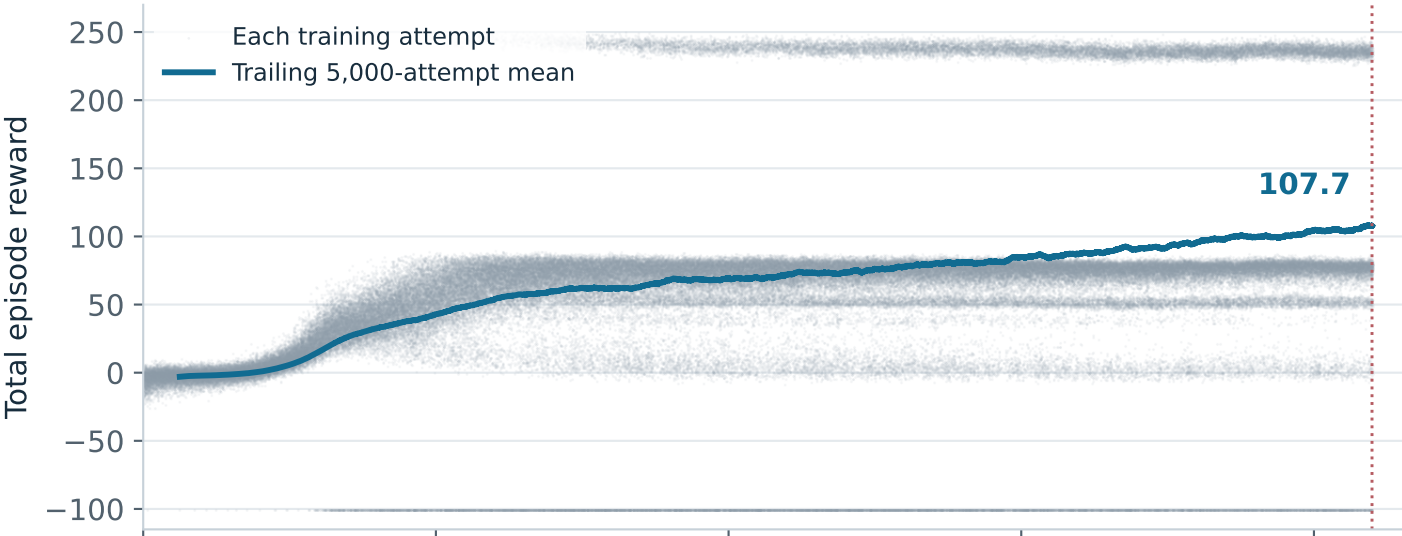


PPO learning before the crash

167,936 recorded attempts | 164 batches of 1,024 | Native 30 Hz, both residuals

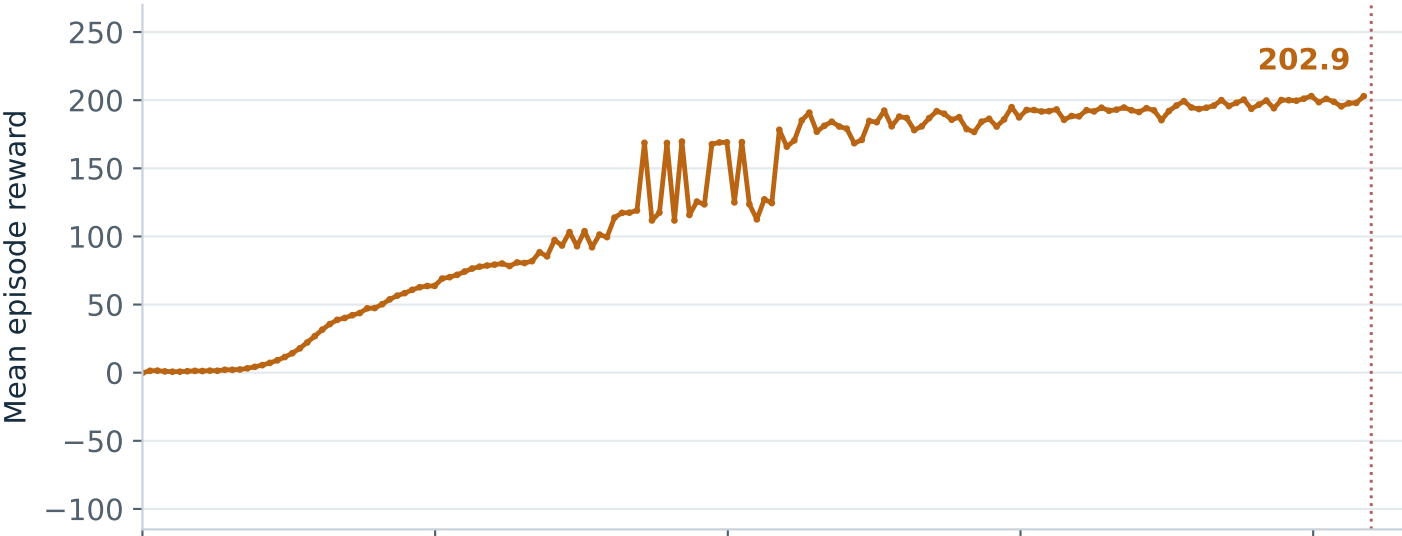
EXPLORATORY TRAINING

Reward per attempt

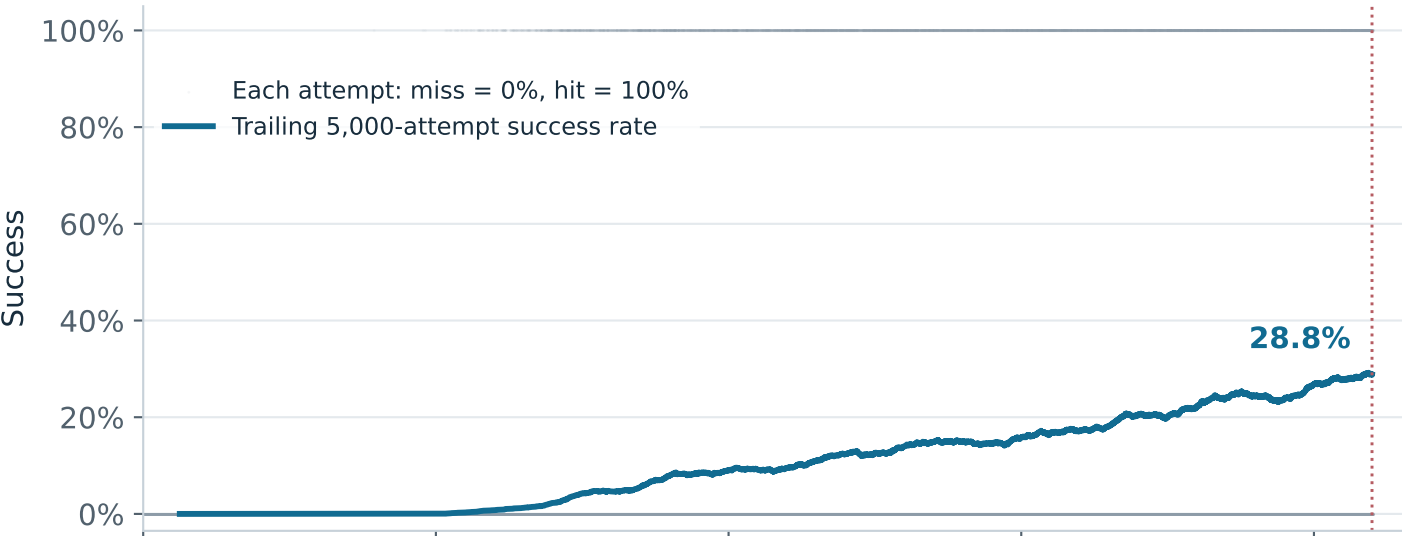


DETERMINISTIC SIMULATED VALIDATION

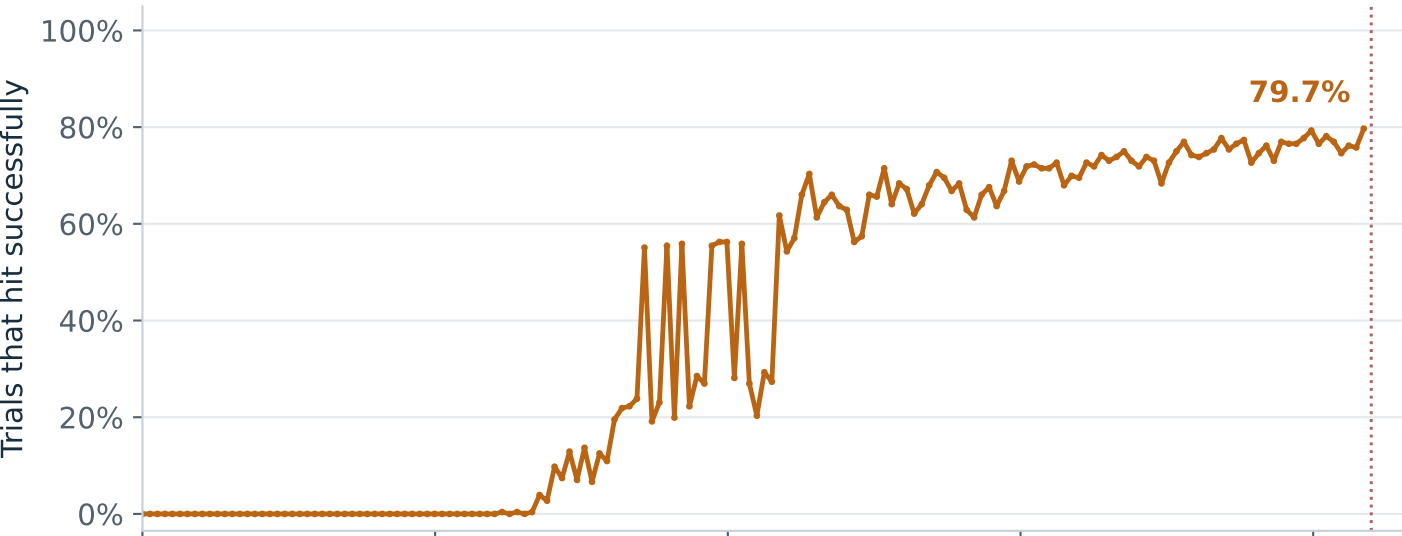
Mean reward of 256 validation trials



Success per attempt and rolling rate



Success rate on the fixed validation scenarios



Grey points are individual training attempts; blue lines average the previous 5,000 attempts. Training uses exploration; validation uses the deterministic policy. Attempts within each parallel batch have arbitrary order. Dotted line: crash at 167,936. Last completed validation: 166,912. Fixed simulation development set.